

Triebel–Lizorkin RKBS Pairs for the Bessel Kernel: A Characterization with Finite Microscopic Exponents

Run fa_banach_001, agent agent_lane_01

2026-07-22

Source Question

Heeringa characterizes all pairs of Bessel potential spaces having the Bessel (Matérn) kernel

$$K_s(x, y) = G_{2s}(x - y)$$

as reproducing kernel [1]. In the future-work paragraph on PDF page 14 of arXiv:2603.26979, the paper asks for the corresponding Triebel–Lizorkin characterization:

Next, Theorem 1 only considers the Bessel potential spaces. These spaces are a special case of the larger family of Triebel–Lizorkin $F_{p,q}^s(\mathbb{R}^d)$: The Bessel potential spaces are isomorph to the special case $q = 2$. It is natural to ask which Triebel–Lizorkin spaces admit K_s as kernel.

In transcription: “The Bessel potential spaces are isomorph to the special case $q = 2$. It is natural to ask which Triebel–Lizorkin spaces admit K_s as kernel.” The repeated letter q in the source denotes the microscopic Triebel–Lizorkin exponent. Below we use p, q for spatial integrability and a, b for the microscopic exponents.

Result and Scope

The main theorem completely answers the question when

$$1 < p, q < \infty, \quad 1 \leq a, b < \infty.$$

Thus the spatial exponents are reflexive and the microscopic exponents are finite; $a = 1$ and $b = 1$ are included. The spatial endpoints and the nonseparable cases $a = \infty$ or $b = \infty$ are not claimed.

We use the inhomogeneous Triebel–Lizorkin spaces on $\mathbb{R}^d$. Put

$$\Lambda^t = (I - \Delta)^{t/2}, \quad \widehat{G_\alpha}(\xi) = (1 + |\xi|^2)^{-\alpha/2}.$$

This defines G_α as a tempered distribution for every real α . In particular $G_0 = \delta_0$, $\Lambda^t G_\alpha = G_{\alpha-t}$, and $\Lambda^{2s} G_{2s} = \delta_0$. Heeringa’s operator $\mathcal{J}_{-2s}$ is our Λ^{2s} .

Idea of the Proof

Three sharp facts fit together.

First, the kernel itself determines the pairing. On Schwartz functions, the two reproducing identities force

$$[g, f] = \langle \Lambda^{2s} f, g \rangle.$$

Since Schwartz functions are dense for finite microscopic exponents, no other continuous pairing can evade this identity. Triebel–Lizorkin duality therefore turns existence of the pairing into the sharp embedding

$$F_{p,a}^{u-2s} \hookrightarrow F_{q',b'}^{-v}.$$

Second, critical Triebel–Lizorkin Sobolev embeddings behave differently according to whether spatial integrability improves. If $p < q'$, the critical embedding is independent of a, b . If $p = q'$, there is no spatial gain and equality requires $a \leq b'$, equivalently $1/a + 1/b \geq 1$. This is the only new boundary clause.

Third, one dyadic calculation gives the exact membership criterion

$$G_\alpha \in F_{r,c}^t \iff t < \alpha - \frac{d}{r'}.$$

Taking $G_0 = \delta_0$ detects bounded point evaluations; taking G_{2s} detects membership of all kernel sections. These three facts give both necessity and sufficiency.

Function-Space Inputs

We record the standard facts used below. For $1 < r < \infty$, $1 \leq c < \infty$, and $z, t \in \mathbb{R}$,

$$(F_{r,c}^z)^* = F_{r',c'}^{-z}, \tag{D}$$

$$\Lambda^t : F_{r,c}^z \longrightarrow F_{r,c}^{z-t} \text{ is an isomorphism.} \tag{L}$$

The duality formula includes $c = 1$, in which case $c' = \infty$. Moreover, $\mathcal{S}(\mathbb{R}^d)$ is dense when $c < \infty$. These are classical Triebel–Lizorkin facts; see Triebel [3, Sections 2.3 and 2.11].

We also use the following exact unweighted embedding theorem, obtained by specializing Theorem 1.2 of Meyries–Veraar [2].

Proposition 1 (Sharp Triebel–Lizorkin embedding). *Let $1 < r_0, r_1 < \infty$, $1 \leq c_0, c_1 \leq \infty$, and $z_0, z_1 \in \mathbb{R}$. Then*

$$F_{r_0,c_0}^{z_0} \hookrightarrow F_{r_1,c_1}^{z_1}$$

if and only if one of the following holds:

1. $r_0 < r_1$ and $z_0 - d/r_0 \geq z_1 - d/r_1$;
2. $r_0 = r_1$, and either $z_0 > z_1$, or $z_0 = z_1$ and $c_0 \leq c_1$.

No such embedding holds when $r_0 > r_1$.

The important feature is that, when $r_0 < r_1$, equality of Sobolev indices imposes no condition on c_0, c_1 .

The Bessel-Distribution Lemma

Lemma 1 (Exact membership). *Let $\alpha, t \in \mathbb{R}$, $1 < r < \infty$, and $1 \leq c \leq \infty$. Then*

$$G_\alpha \in F_{r,c}^t(\mathbb{R}^d) \iff t < \alpha - \frac{d}{r'}.$$

Proof. By the lifting property, it is enough to prove

$$G_\beta \in F_{r,c}^0 \iff \beta > \frac{d}{r'}, \quad \beta = \alpha - t.$$

Suppose first that $\beta > d/r'$. Choose $\varepsilon > 0$ such that $\beta - \varepsilon > d/r'$. The exact local integrability criterion for Bessel kernels gives

$$\Lambda^\varepsilon G_\beta = G_{\beta-\varepsilon} \in L^r(\mathbb{R}^d);$$

see, for example, Lemma 1 of [1]. Hence $G_\beta \in H^{\varepsilon,r} = F_{r,2}^\varepsilon$. A strict loss of smoothness embeds this space into $F_{r,c}^0$, for every $1 \leq c \leq \infty$.

For necessity, fix a nonzero smooth annular cutoff φ from an inhomogeneous Littlewood–Paley decomposition and write Δ_j for the associated high-frequency blocks. For $j \geq 1$, scaling gives

$$\Delta_j G_\beta(x) = 2^{j(d-\beta)} h_j(2^j x), \quad h_j = \mathcal{F}^{-1} \left[\varphi(\eta) (2^{-2j} + |\eta|^2)^{-\beta/2} \right]. \quad (1)$$

Because φ is supported away from zero, $h_j \rightarrow h = \mathcal{F}^{-1}[\varphi(\eta)|\eta|^{-\beta}]$ in the Schwartz topology, and $h \neq 0$. Therefore

$$\|\Delta_j G_\beta\|_{L^r} \asymp 2^{j(d/r' - \beta)} \quad (j \text{ large}). \quad (2)$$

If $c < \infty$, the elementary embedding

$$F_{r,c}^0 \hookrightarrow B_{r,\max(r,c)}^0$$

and (2) rule out $\beta \leq d/r'$: the block norms would not lie in the required finite $\ell^{\max(r,c)}$ space.

For $c = \infty$, the same Besov embedding rules out $\beta < d/r'$. At equality, (1) becomes

$$\Delta_j G_\beta(x) = 2^{jd/r} h_j(2^j x).$$

Choose a ball B , centered away from the origin, on which $|h| \geq 2\kappa > 0$. Uniform convergence gives $|h_j| \geq \kappa$ on B for all large j . Passing to indices $j_n = j_0 + nL$ with L fixed and sufficiently large makes the annuli $E_n = 2^{-j_n} B$ pairwise disjoint. On E_n ,

$$\sup_j |\Delta_j G_\beta(x)| \geq \kappa 2^{j_n d/r}.$$

Consequently each E_n contributes at least

$$\kappa^r 2^{j_n d} |E_n| = \kappa^r |B|$$

to the r -th power of the $F_{r,\infty}^0$ norm. Summation over the disjoint annuli diverges. Equality is therefore impossible, completing the proof. $\square$

Remark 1. *The case $\alpha = 0$ says*

$$\delta_0 \in F_{r,c}^t \iff t < -d/r'.$$

The strict endpoint is essential below.

Main Characterization

Theorem 1 (Finite-microscopic Triebel–Lizorkin characterization). *Let $d \in \mathbb{N}$, $s > 0$, $u, v \in \mathbb{R}$, $1 < p, q < \infty$, and $1 \leq a, b < \infty$. The spaces*

$$X = F_{p,a}^u(\mathbb{R}^d), \quad Y = F_{q,b}^v(\mathbb{R}^d)$$

form an RKBS pair, in the sense of [1], with kernel $K_s(x, y) = G_{2s}(x - y)$ if and only if

$$\frac{1}{p} + \frac{1}{q} \geq 1, \tag{3}$$

$$\frac{d}{p} < u < 2s - \frac{d}{p'}, \tag{4}$$

$$\frac{d}{q} < v < 2s - \frac{d}{q'}, \tag{5}$$

$$u + v \geq 2s + d \left(\frac{1}{p} + \frac{1}{q} - 1 \right). \tag{6}$$

In the doubly critical case

$$\frac{1}{p} + \frac{1}{q} = 1, \quad u + v = 2s, \tag{7}$$

one must additionally have

$$\frac{1}{a} + \frac{1}{b} \geq 1. \tag{8}$$

There is no microscopic restriction at equality in (6) when $1/p + 1/q > 1$.

Necessity. Suppose X, Y form an RKBS pair with this kernel.

Point evaluations. Evaluation at zero is bounded on X . On the dense Schwartz subspace, this functional is the distribution δ_0 ; hence, by (D),

$$\delta_0 \in X^* = F_{p',a'}^{-u}.$$

Lemma 1, with $\alpha = 0$, gives $-u < -d/p$, that is, $u > d/p$. The same argument on Y gives $v > d/q$.

Kernel sections. The RKBS-pair definition requires $K_s(\cdot, y) \in X$ and $K_s(x, \cdot) \in Y$. Translations are isometries, so Lemma 1, now with $\alpha = 2s$, gives

$$u < 2s - d/p', \quad v < 2s - d/q'.$$

This proves the strict bounds in (4)–(5).

The pairing is forced. Write $[g, f]$ for the continuous pairing on $Y \times X$. For $f, \psi \in \mathcal{S}(\mathbb{R}^d)$, the first reproducing identity and continuity of the pairing give

$$\begin{aligned} \int_{\mathbb{R}^d} f(x) \psi(x) dx &= \int_{\mathbb{R}^d} [K_s(x, \cdot), f] \psi(x) dx \\ &= \left[\int_{\mathbb{R}^d} \psi(x) G_{2s}(x - \cdot) dx, f \right] \\ &= [\Lambda^{-2s} \psi, f]. \end{aligned}$$

The Bochner integral is legitimate because translations are strongly continuous and have constant Y -norm. Since Λ^{-2s} is an automorphism of $\mathcal{S}$, putting $g = \Lambda^{-2s} \psi$ yields

$$[g, f] = \langle f, \Lambda^{2s} g \rangle = \langle \Lambda^{2s} f, g \rangle \quad (f, g \in \mathcal{S}). \tag{9}$$

Density and continuity therefore force

$$\Lambda^{2s} : X \longrightarrow Y^* = F_{q',b'}^{-v}$$

to be bounded. By lifting, this is precisely

$$F_{p,a}^{u-2s} \hookrightarrow F_{q',b'}^{-v}. \quad (10)$$

Proposition 1 first forces $p \leq q'$, which is (3). If $p < q'$, (10) is equivalent to

$$u - 2s - \frac{d}{p} \geq -v - \frac{d}{q'},$$

namely (6); equality permits arbitrary a, b . If $p = q'$, the embedding holds exactly when $u + v > 2s$, or when $u + v = 2s$ and $a \leq b'$. The latter inequality is equivalent to (8). This proves every necessary condition. $\square$

Sufficiency. Assume (3)–(6), together with (8) when (7) holds.

The strict lower bounds imply the standard embeddings

$$X \hookrightarrow C_b(\mathbb{R}^d), \quad Y \hookrightarrow C_b(\mathbb{R}^d).$$

For completeness, combine $F_{p,a}^u \hookrightarrow B_{p,\max(p,a)}^u$ with Bernstein's inequality: the factor $2^{-j(u-d/p)}$ makes the Littlewood–Paley sum converge uniformly. Thus both spaces are Banach spaces of bounded continuous functions with bounded evaluations.

The strict upper bounds and Lemma 1 give all kernel-section inclusions

$$G_{2s}(\cdot - y) \in X, \quad G_{2s}(x - \cdot) \in Y.$$

The remaining parameter conditions and Proposition 1 give (10). Define

$$[g, f] = \langle \Lambda^{2s} f, g \rangle_{Y^*, Y}. \quad (11)$$

This pairing is continuous. It separates X and Y : the operator Λ^{2s} is injective on tempered distributions, and Schwartz tests separate every nonzero tempered distribution. Self-adjointness gives the same conclusion in the other variable.

Finally, $\Lambda^{2s} G_{2s}(\cdot - y) = \delta_y$. Hence

$$[g, K_s(\cdot, y)] = \langle \delta_y, g \rangle = g(y).$$

Using self-adjointness in (11) also gives

$$[K_s(x, \cdot), f] = \langle \Lambda^{2s} f, G_{2s}(x - \cdot) \rangle = \langle f, \delta_x \rangle = f(x).$$

Both reproducing identities hold, so X, Y form the asserted RKBS pair. $\square$

Verification Notes

The theorem passes the following independent consistency checks.

- Setting $a = b = 2$ recovers exactly the part of Heeringa's main theorem with $1 < p, q < \infty$.

- On the Hölder face $p = q'$, equality $u + v = 2s$ reduces the pairing condition to the sequence-space threshold $a \leq b'$, equivalently $1/a + 1/b \geq 1$.
- The characterization is invariant under interchanging (u, p, a) and (v, q, b) .
- Combining $u > d/p$ with $u < 2s - d/p'$ forces $2s > d$, so a separate lower bound on s is unnecessary.
- All strict endpoint claims for evaluations and kernel sections are instances of Lemma 1; no endpoint is inferred from a one-sided embedding alone.

No numerical computation is used as proof evidence. The only nonstandard lemma is proved explicitly by high-frequency scaling. The decisive external input, Proposition 1, is an if-and-only-if embedding theorem and its source PDF is included with this packet.

Limitations and Novelty Check

The result does not cover $p = 1$ or $q = 1$, where Hardy/BMO-type duals replace the simple formula (D). It also excludes $a = \infty$ or $b = \infty$: Schwartz functions are then not dense, so the reproducing identities need not force a pairing on the entire space by the present argument. Exponents below one are quasi-Banach and fall outside the source definition of an RKBS pair.

A bounded novelty search was performed on 2026-07-22. The run registry, solution, attempt, and proof-gap indexes and the local 2000–2026 arXiv title/abstract corpus were searched for arXiv:2603.26979 and combinations of “RKBS,” “reproducing kernel Banach,” “Triebel–Lizorkin,” “Bessel kernel,” and “Matérn kernel.” Web/arXiv searches used the exact source title and the phrases “reproducing kernel Banach + Triebel–Lizorkin,” “Bessel potential kernel + Triebel–Lizorkin + RKBS,” and “Matérn kernel + Triebel–Lizorkin.” They returned the March 2026 source paper but no later or parallel characterization. The nearest result found concerns embedding Triebel–Lizorkin spaces into some RKHS on bounded domains, a different question from identifying pairs with this fixed kernel. The search supports novelty but is not a claim of exhaustive global priority.

Human Review Recommendation

Review as a strong partial solution. The highest-value checks are the $c = \infty$ endpoint in Lemma 1, the duality formula when $a = 1$ or $b = 1$, the Bochner-integral step in (9), and the split between $p < q'$ and $p = q'$ in Proposition 1. If these pass, the packet gives a complete and sharp theorem throughout the stated finite-microscopic range; it should not be labeled a full solution of all Triebel–Lizorkin endpoints.

References

- [1] T. J. Heeringa, *Characterization of the reproducing structure of the Bessel potential spaces beyond $p = 2$* , arXiv:2603.26979 (2026).
- [2] M. Meyries and M. Veraar, *Sharp embedding results for spaces of smooth functions with power weights*, arXiv:1112.5388 (2012).
- [3] H. Triebel, *Theory of Function Spaces*, Monographs in Mathematics 78, Birkhäuser, Basel, 1983.